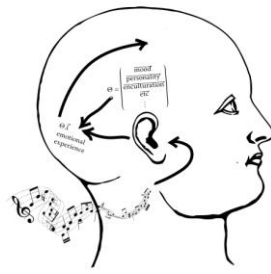


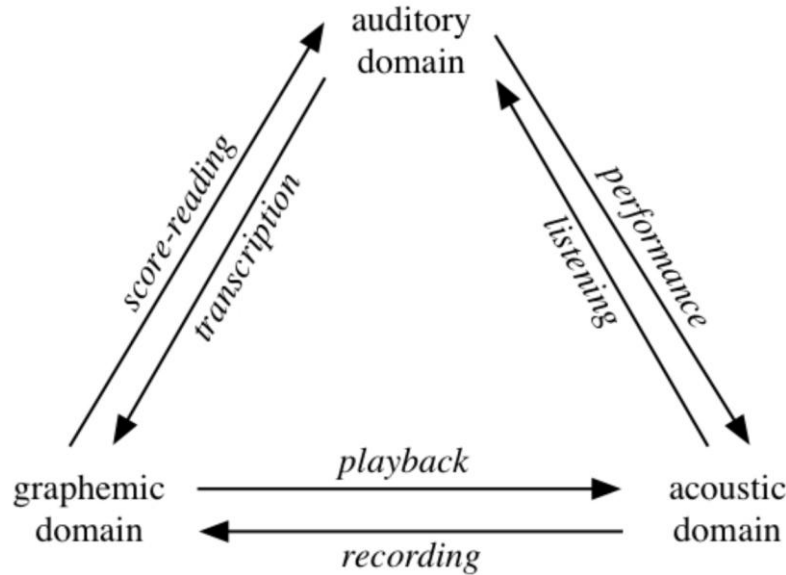
Music representations. Symbolic music formats and datasets.

Plan

- Music representations
- Symbolic music formats
- Datasets and their organization
- How to read plots and figures



“Neither music nor language can be studied as pure surface forms, because the cognition of both produces information which is not contained in the surface form”

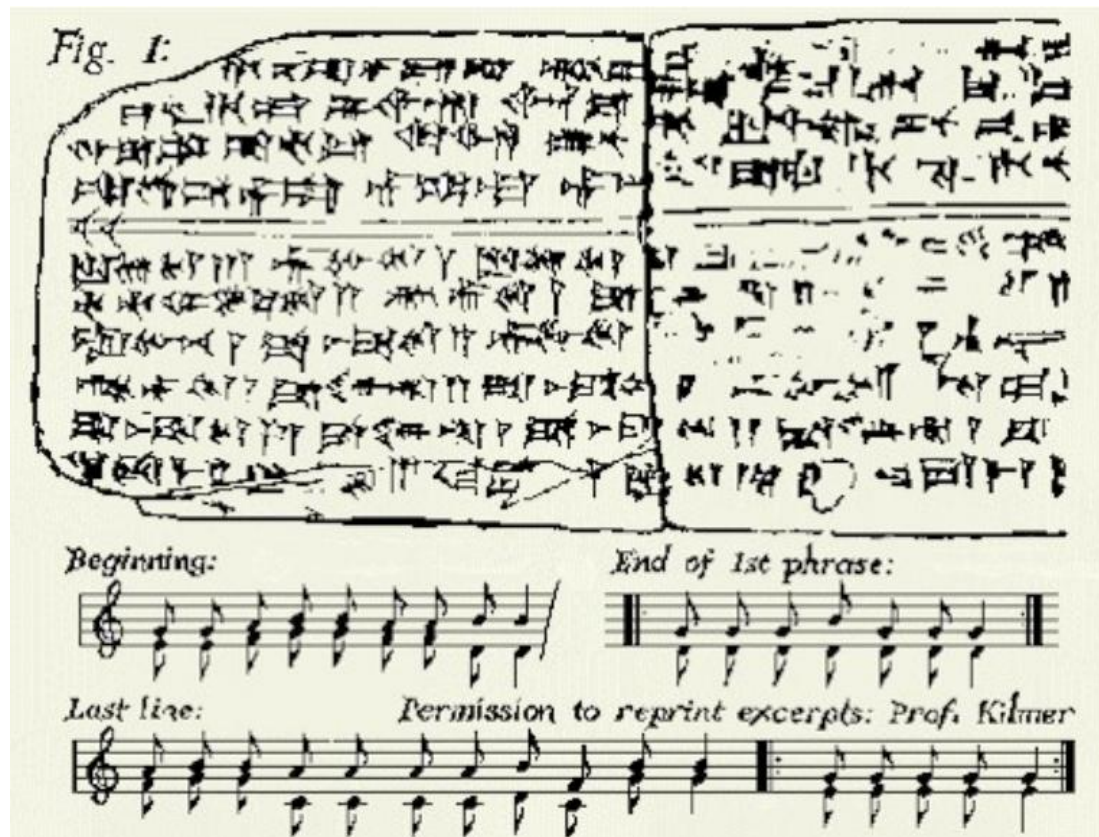


*On the non-existence of music: Why music theory is a figment of the imagination. G. A. Wiggins et al. (2010), Musicae Scientiae, Discussion Forum, 5, pp. 231-255 *

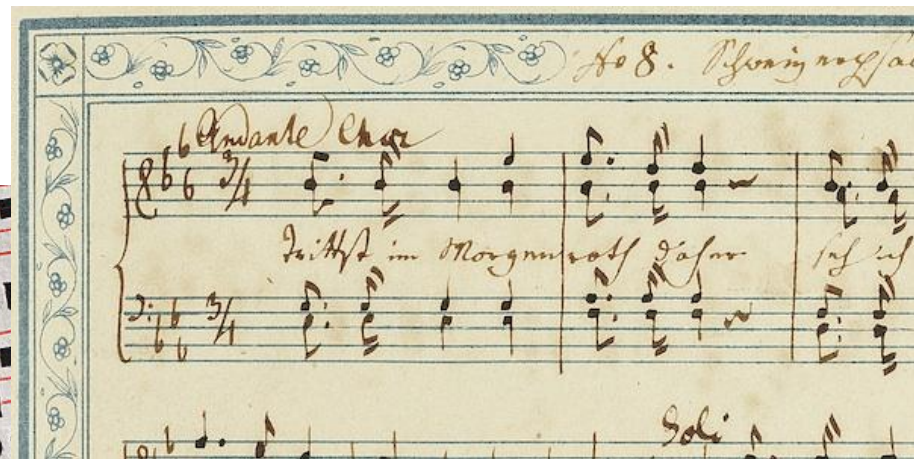
Babbitt, M. (1965). The use of computers in musicological research. Perspectives of New Music, 3(2), 74-83.

Symbolic music representations

aka graphemic domain

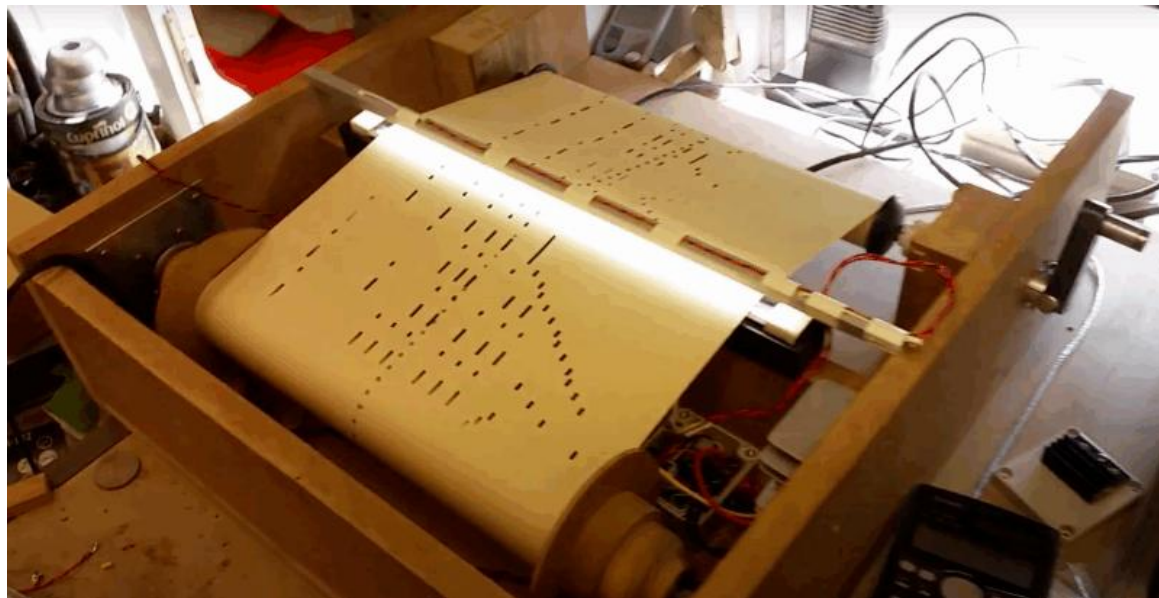


Cuneiform tablets (~1400 BC)

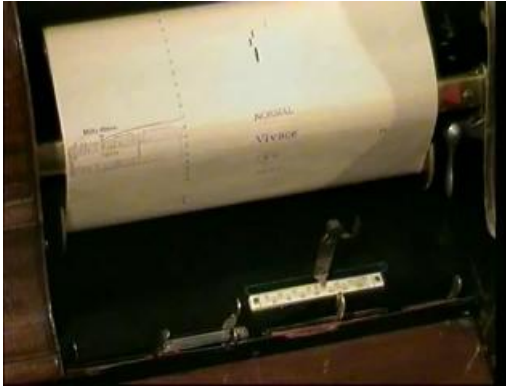


Byzantine, Medieval Europe, modern music notation

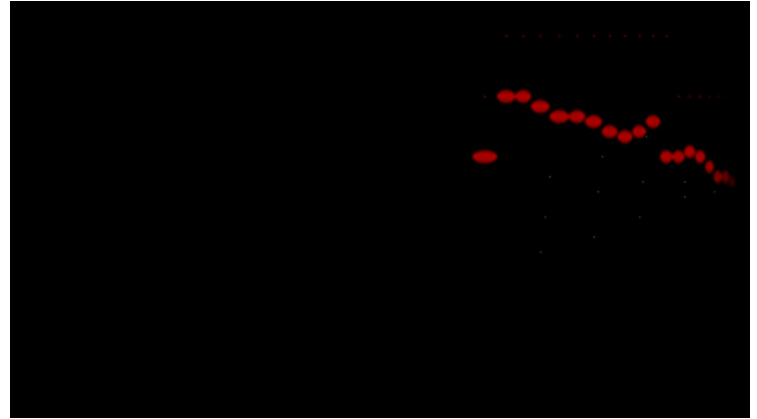
Piano roll



Piano roll



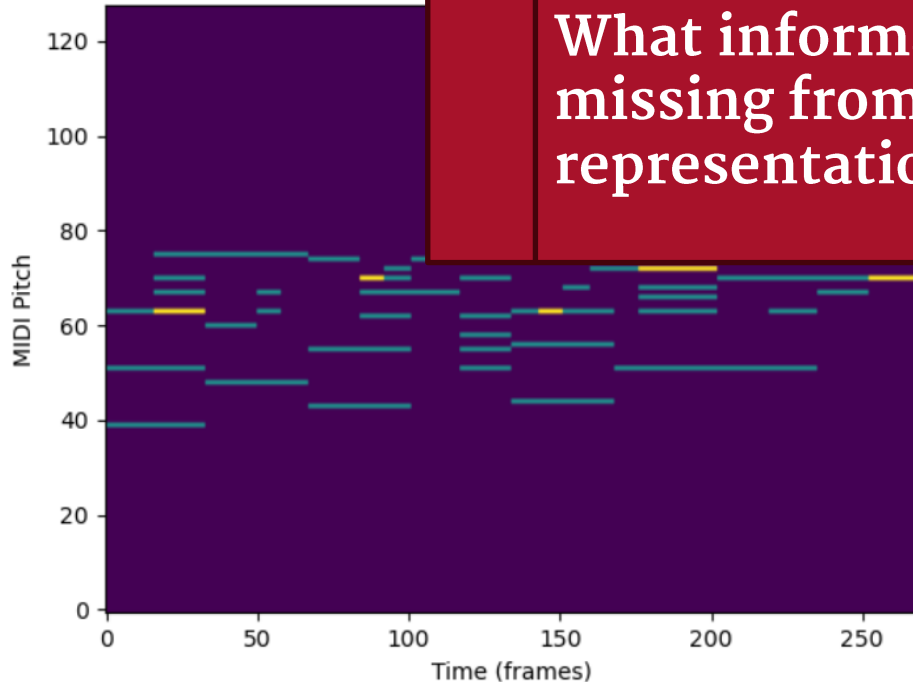
<https://www.youtube.com/watch?v=2A6ZXZwl3nA>



<https://www.youtube.com/watch?v=mShl7KDYPzs>

Piano roll

What information is missing from this representation?



```
# a library to read, write MIDI files  
#for Python
```

```
import pretty_midi
```

```
pm = pretty_midi.PrettyMIDI('file.mid')
```

```
#fs parameter - frames per second
```

```
piano_roll = pm.get_piano_roll(fs=25)
```

Can you recognize this piece?

$\text{♩} = 89$

Soprano

Piano

Red. _____



MIDI format

- MIDI: Musical Instrument Digital Interface
- Standardized in 1983
- NoteOn / NoteOff events with delta timing
- Velocity value for events (corresponding to loudness)
- Program (instrument)
- Control changes (pedals, modulation)

MetaMessage('track_name', name='Soprano', time=0)

MetaMessage('time_signature', numerator=4,
denominator=4, clocks_per_click=24,
notated 32nd notes per beat=8, time=0)

MetaMessage('key_signature', key='Eb', time=0)

MetaMessage('set_tempo', tempo=674157, time=0)

note_on channel=1 note=63 velocity=80 time=0

MetaMessage('lyrics', text='Some', time=0)

note_off channel=1 note=63 time=959

note_on channel=1 note=75 velocity=80 time=1

MetaMessage('lyrics', text='where ', time=0)

note_off channel=1 note=75 time=959

note_on channel=1 note=74 velocity=80 time=1

MetaMessage('lyrics', text='o', time=0)

note_off channel=1 note=74 time=479



MIDI: four bars of melody

```
MetaMessage('track_name', name='Soprano',
MetaMessage('time_signature', numerator=4,
denominator=4, clocks_per_click=24,
notated_32nd_notes_per_beat=8, time=0)
MetaMessage('key_signature', key='Eb', time=
MetaMessage('set_tempo', tempo=674157, tin
note_on channel=1 note=63 velocity=80 time=
MetaMessage('lyrics', text='Some', time=0)
note_off channel=1 note=63 time=959
note_on channel=1 note=75 velocity=80 time=
MetaMessage('lyrics', text='where ', time=0)
note_off channel=1 note=75 time=959
note_on channel=1 note=74 velocity=80 time=1
MetaMessage('lyrics', text='o', time=0)
note_off channel=1 note=74 time=479
```

```
import mido
```

```
mid = mido.MidiFile('file.mid')
```

```
for i, track in enumerate(mid.tracks):
```

```
    print(f'--- Track {i}: {track.name} ---')
```

```
    for msg in track:
        print(msg)
```



How MIDI files work

```
MetaMessage('track_name', name='Soprano', time=0)
MetaMessage('time_signature', numerator=4,
denominator=4, clocks_per_click=24,
notated_32nd_notes_per_beat=8, time=0)
MetaMessage('key_signature', key='Eb', time=0)
MetaMessage('set_tempo', tempo=674157, time=0)
```

```
note_on channel=1 note=63 velocity=80 time=0
```

```
MetaMessage('lyrics', text='Some', time=0)
```

```
note_off channel=1 note=63 time=959
```

```
note_on channel=1 note=75 velocity=80 time=1
```

```
MetaMessage('lyrics', text='where ', time=0)
```

```
note_off channel=1 note=75 time=959
```

```
note_on channel=1 note=74 velocity=80 time=1
```

```
MetaMessage('lyrics', text='o', time=0)
```

```
note_off channel=1 note=74 time=479
```

A musical score for the song "Some-where over the rainbow". The score is written for Soprano (treble clef) and Piano (grand staff). The tempo is marked as ♩ = 89. The key signature is three flats (B-flat, E-flat, A-flat) and the time signature is 4/4. The lyrics are "Some-where over the rainbow". The Soprano part has three notes highlighted with red boxes: "Some", "where", and "o". The Piano part has a "Red." marking at the bottom.

How MIDIs work

note_on channel=0 note=63 velocity=80 time=0

note_off channel=0 note=63 time=239

note_on channel=0 note=63 velocity=80 time=241

note_off channel=0 note=63 time=479



How MIDI works

MEI

- MEI (Music Encoding Initiative)
- Represents music notation using XML markup
- Developed in the early 21st century
- As compared to musicXML, caters more to digital musicology than music typesetting

MEI

```

<measure xml:id="m2rsmfx" n="1">
  <staff xml:id="m1s1" n="1">
    <layer xml:id="m1s1l1" n="1">
      <note xml:id="n1vcmnhv" dur="2" pname="e" oct="4">
        <verse xml:id="v134emuv" n="1">
          <syl con="d" wordpos="i">Some</syl>
        </verse>
        <accid accid.ges="f" />
      </note>
      <note xml:id="n19ejbnu" dur="2" pname="e" oct="5">
        <verse xml:id="v1knsqd0" n="1">
          <syl wordpos="t">where</syl>
        </verse>
        <accid accid.ges="f" />
      </note>
    </layer>
  </staff>

```

The image shows a musical score for the song "Some-where over the rainbow". The tempo is marked as ♩ = 89. The key signature has two flats (B-flat and E-flat), and the time signature is 4/4. The score consists of two staves: a vocal melody on a treble clef staff and a piano accompaniment on a grand staff (treble and bass clefs). The lyrics "Some - where o - ver the rain-bow" are written below the vocal staff. A red rectangular box highlights the first two measures of the vocal line, which correspond to the words "Some" and "where". In the piano accompaniment, the first measure of the vocal line is marked with a "Red." and a bracket, indicating a reduction or specific annotation.

ABC notation

- Created in the 1990s by Chris Walshaw
- A simple text based notation to encode folk tunes
- Overtime evolved to support multiple voices, more metadata
- Human-readable easy format
- Large corpora of folk music
 - <https://abcnotation.com/tunes>
 - folktunefinder.com

Header

ABC notation

Text

```
X:1
T:Somewhere over the rainbow
C:Composer
%%score 1 2
L:1/4
Q:1/4=89
M:4/4
I:linebreak $
K:Eb
```

```
V:1 treble nm="Soprano" snm="S."
V:2 treble nm="Piano" snm="Pno."
V:1
E2 e2 | d B/c/ d e | E2 c2 | B4 || %4
w: Some- where | o- ver the rain- bow | way up | high|
V:2
[EGBe]2 [EG]2 | [DGB]2 [DB]2 | [EAc]2 [E_GAc]2 | [EGB]2 z z
|| %4
```

The image displays a musical score for the song 'Somewhere over the rainbow'. It features two staves: a Soprano staff and a Piano staff. The Soprano staff is in treble clef with a key signature of three flats (B-flat, E-flat, A-flat) and a 4/4 time signature. The tempo is marked as quarter note = 89. The lyrics are written below the notes: 'Some - where o - ver the rain - bow way up high'. The Piano staff is also in treble clef with the same key signature and time signature, providing harmonic accompaniment with chords and single notes.

Header

ABC notation

Text

```
X:1
T:Somewhere over the rainbow
C:Composer
%%score 1 2
L:1/4
Q:1/4=89
M:4/4
I:linebreak $
K:Eb
```

```
V:1 treble nm="Soprano" snm="S."
```

```
V:2 treble nm="Piano" snm="Pno."
```

```
V:1
```

```
E2 e2 d B/c/ d e | E2 c2 | B4 ] ] %4
```

```
w: Some- where | o- ver the rain- bow | way up | high|
```

```
V:2
```

```
[EGB]2 [EG]2 | [DGB]2 [DB]2 | [EAc]2 [E_GAc]2 | [EGB]2 z z  
]] %4
```

A musical score for the song 'Somewhere over the rainbow'. The score is written for Soprano and Piano. The Soprano part is in treble clef, 4/4 time, with a key signature of two flats (Bb and Eb). The Piano part is also in treble clef, 4/4 time, with the same key signature. The lyrics are: 'Some - where o - ver the rain - bow way up high'. The first measure of the Soprano part, containing the notes E2 and e2, is highlighted with a red box. The first measure of the Piano part, containing the chords [EGB]2 and [EG]2, is also highlighted with a red box. The tempo is marked as 89.

Lead sheet

Eb Cm Gm Eb7 Ab Ab7
Somewhere over the rainbow, way up high

The image shows a musical score for the song 'Somewhere over the rainbow, way up high'. It is a lead sheet, meaning it contains the melody and lyrics for the vocal part, and the harmonic accompaniment for the piano. The score is written in E-flat major (three flats) and 4/4 time. The tempo is marked as quarter note = 89. The vocal part is for Soprano, and the piano part is for Piano. The lyrics are 'Some - where o - ver the rain - bow way up high'. The piano accompaniment consists of chords and single notes that support the melody.

Soprano

$\text{♩} = 89$

Some - where o - ver the rain - bow way up high

Piano

Comparison of formats

Feature	Piano roll	MIDI	MEI	ABC	Lead sheet
Encodes pitch	✓	✓	✓	✓	?
Encodes rhythm	✓	✓	✓	✓	✗
Shows human-readable notation	✗	✗	✓	✓	✓
Meter notation	✗	✓	✓	✓	✗
Dynamics notation	✗	✓	✓	✗	✗
Can store performance information	✗	✓	✗	✗	✗
Lyrics	✗	✓	✓	✓	✓
Editorial markup	✗	✗	✓	✗	✗

**What kind of questions
can we answer using data
in various formats?**

****kern (Humdrum framework)**

- David Huron, c. 1990
 - represent basic or core information for period-of-common-practice Western music
- Provides limited capabilities for representing accidentals, articulation, ornamentation, ties, slurs, phrasing, barlines, stem-direction and beaming
- Designed to facilitate analytic applications rather than music printing or sound generation
- Organized in spines (columns)

****kern (Humdrum framework)**

- David Huron, c. 1990
 - represent basic or core information for period-of-common-practice music
- Provides limited capabilities for representing accidentals, articulation, phrasing, barlines, etc.
- Designed to facilitate music printing or score generation
- Organized in spines (columns)

**Bass	**Tenor	**Alto	**Soprano
B1	T1	A1	S1
.	.	.	S2
B2	T2	A2	S3
.	.	.	S4
B3	T3	A3	S5
etc.	etc.	etc.	etc.
*_	*_	*_	*_

- Music representation and control
 - Csound
 - Music macro language
- DARMS
- MusicTeX
- Braille music notation

Digital corpora

Digital corpus vs digital library

DIGITAL LIBRARY

Has an archival purpose

Includes any available materials for preservation and storage

Diverse formats, medium

Organized and described differently depending on an object

Inconsistent metadata

DIGITAL CORPUS

Created for analytical purpose

Explicit inclusion criteria

Intentionally designed to answer certain questions and to represent a certain phenomenon

Has consistent metadata, data format, detailed documentation, data provenance

Understanding data: scope and limitations

- What purpose does our dataset serve?
- What are the selection criteria (what is excluded)?
- How does the data format and quality affect what is possible to do with it?
- Are data contributors aware of the data usage?

Understanding data: scope and limitations

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- Are data contributors aware of the data u



Data always encodes existing biases. Are we perpetuating them?

- The motivation for dataset creation
- The composition of the dataset
- The data collection process
- The preprocessing of the data
- The distribution of the data
- The maintenance of the data
- The legal and ethical considerations

Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumeé III, Kate Crawford.

Datasheets for Datasets, 2018 <https://arxiv.org/abs/1803.09010?context=cs>

FAIR data principles

- Findable 
 - DOI, github
- Accessible 
 - Online access (can be through authentication)
- Interoperable 
 - Dublin core metadata, data management plan
- Reusable 
 - Clear licensing and documentation

Dublin Core

- International metadata standard (Dublin Core Metadata Element Set)
- Includes 15 metadata terms
- Uniformly describe a resource
- Improve SEO

https://nsteffell.github.io/dublin_core_generator/generator_nq.html

Exercise: examine
existing datasets

Checklist:

- What is included and excluded:
 - Historically
 - Geographically
 - How does the data format dictate what is feasible?
- Are data owners and producers aware of the data usage? Could the dataset be used unethically? How?
- How was the data cleaned? Could it influence results?
- How could this data be made future-proof?